Q1. What is NumPy? Why should we use it?

Ans:

NumPy stands for Numerical Python. It is a Python library used to perform mathematical and numerical operations on large amounts of data.

It provides a powerful object called an array, which allows us to store and manage multiple values efficiently.

Why should we use NumPy?

1. Fast Execution: NumPy performs calculations faster than normal Python lists.
2. Less Memory: It uses less memory to store large amounts of data.
3. Mathematical Operations: It helps perform addition, subtraction, multiplication, and division easily.
4. Multidimensional Arrays: It supports 1D, 2D, and 3D arrays.
5. Data Analysis: It is useful in data science, machine learning, and data analysis.

Q2. Write the steps to create 2D and 3D arrays with output.

Ans:

A. Creating a 2D Array

Step 1: Import the NumPy library.

Step 2: Create a 2D array using np.array().

Step 3: Print the array.

Program:

```
import numpy as np


arr = np.array([[1, 2, 3],
                [4, 5, 6]])


print(arr)
```

Output:

```
[[1 2 3]
 [4 5 6]]
```

Explanation: A 2D array contains rows and columns. In this example, there are 2 rows and 3 columns.

B. Creating a 3D Array

Step 1: Import the NumPy library.

Step 2: Create a 3D array using np.array().

Step 3: Print the array.

Program:

```
import numpy as np


arr = np.array([[[1, 2, 3],
                 [4, 5, 6]],


                [[7, 8, 9],
                 [10, 11, 12]]])


print(arr)
```

Output:

```
[[[ 1  2  3]
  [ 4  5  6]]

 [[ 7  8  9]
  [10 11 12]]]
```

Explanation: A 3D array contains multiple 2D arrays. In this example, it has 2 matrices, each containing 2 rows and 3 columns.